

System Integration & Architecture

00 – OVERVIEW

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4th Level



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Introduction

- Many systems are built to easy, improve and transform organizations.
- Some organizations have many departments which run systems which are independent of each other.
- And systems built sometimes, may not have an abstract view (architecture) which leads to failure of system interoperability.
- There is need to have architectural view of the system as a priority to help in the design to avoid the likeliness of system failure.

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- Besides after the system has been designed and developed in consideration of the size of the organization, i.e. most especially when the organization is large, need is required to integrate such systems to ensure flexibility, Speed, Cost, Standardization, Data integrity, reliability and robustness.
- This can help Information Technology (IT), energy, and financial services industry among others to have an easy to use integrated system.

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What students need to know

- Systems Integration (SI) process, approaches, drivers, tools and techniques required for successful SI, critical success factors, and best practices.
- The course focuses on how a proposed system will be integrated with other existing or planned systems.
- It addresses the System Integration problem using architectures as the basis and then addresses the evaluation of the architectures in terms of the capabilities they provide.

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What students need to learn

- The theory and practice of business process integration, legacy integration, new systems integration, business-to-business integration, integration of commercial-off-the-shelf (COTS) products, interface control and management, testing, integrated program management, integrated Business Continuity Planning (BCP).

Aims

- To provide the students an understanding of the technical and business process issues involved in systems integration.

Learning outcomes

- On completion of this course, the students will be able to:
 - Identify integration issues upfront in the process of System Integration and should be able to identify the best practices that ensure successful System Integration.
 - Have an understanding of the technical and business process issues involved in systems integration.

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Indicative content

- The System of Systems Integration Problem
 - Human, Organizational, Societal Cultural, Economic, and Technological aspects;
 - Processes, approaches, drivers, tools and techniques required for successful SI, critical success factors, and best practices in Systems Integration;
 - The Role of Architectures in Systems Integration;
 - Integration in a System of Systems and a Federation 60 of Systems;
 - Model Based Architecture, Design, and Integration;
 - Systems of Systems Interoperability;
 - Evaluation of architectures;
 - Measures of Performance and Effectiveness;

Indicative content

Assessment of System Capabilities;

Analysis of Alternatives;

Case studies and examples from the Information Technology (IT), energy, and financial services industry to illustrate the concepts discussed.

- The theory and practice of business process integration, legacy integration, new systems integration, business-to-business integration, integration of commercial-off-the-shelf (COTS) products, interface control and management, testing, integrated program management, integrated Business Continuity Planning (BCP). Specific focus will be given to issues of interface integration and interoperability of systems.

Key terminologies in this course

- Various key terminologies shall be used throughout this course as follows
- System
- Systems thinking
- System Integration
- System Architecture
- Project

System

- An array of components designed to accomplish a particular objective according to plan. Many sub-systems may be designed which later on are combined together to form a system which is intended to achieve a specific objective which may be set by the Project manager.

Systems thinking

- Is a way of understanding an entity in terms of its purpose, as three steps
 - The three major steps followed in systems thinking
1. Identify a containing whole (system), of which the thing to be explained is a part.
 2. Explain the behavior or properties of the containing whole.
 3. Explain the behavior or properties of the thing to be explained in terms of its *role(s) or function(s) within its containing whole*

(Ackoff, 1981)

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Is the combination of ^{من العناصر} inter-related elements to achieve a common objective (s).



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System Architecture

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- The architecture of a system defines its high-level structure, exposing its gross organization as a collection of interacting components.
- Elements needed to model a software architecture include:
 - Components, Connectors, Systems, Properties and Styles.

What is a project?

- From the key terms described above, a system developer and architects cannot do anything without first establishing various projects. These projects may be new or existing.
- So it is inevitable to first understand what a project is, factors that influence the project, who the owners are and many more as discussed below.

What Is a Project?

- A project is a temporary endeavor undertaken to accomplish a unique product or service
- Attributes of projects
 - unique purpose
 - temporary
 - require resources, often from various areas
 - should have a primary sponsor ^{محمول} and/or customer
 - involve uncertainty

Where do information Systems Projects Originate (Sources of Projects)?

New or changed IS development projects come from **problems**, **opportunities**, and **directives** and are always subject to one or more **constraints**.

Problems – may either be current, suspected, or anticipated. Problems are undesirable situations that prevent the business from fully achieving its purpose, goals, and objectives (users discovering real problems with existing IS).

An **Opportunity** – is a chance to improve the business even in the absence of specific problems. This means that the business is hoping to create a system that will help it with increasing its revenue, profit, or services, or decreasing its costs.

A **Directive** – is a new requirement that is imposed by management, government, or some external influence i.e. are mandates that come from either an internal or external source of the business.

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Projects Cannot Be Run in Isolation

- Projects must operate in a broad organizational environment
- Project managers need to take a holistic or systems view of a project and understand how it is situated within the larger organization

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Stakeholders

- Stakeholders are the people involved in or affected by project activities

- Stakeholders include

ممول • the project sponsor and project team

داعی • support staff

عملاء • customers

مستخدمين • users

موردين • suppliers

کسانی که منافع - یا ضرر - دارند • opponents to the project

Importance of Stakeholders

- Project managers must take time to identify, understand, and manage relationships with all project stakeholders
- Using the four frames of organizations can help meet stakeholder needs and expectations

• Senior executives are very important stakeholders

Table 2-2. What Helps Projects Succeed?

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According to the Standish Group's report "CHAOS 2001: A Recipe for Success," the following items help IT projects succeed, in order of importance:

Executive support

- User involvement
- Experienced project manager
- Clear business objectives
- Minimized scope
- Standard software infrastructure
- Firm basic requirements
- Formal methodology
- Reliable estimates

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Understanding Organizations

We can analyze a formal organization using the following 4 (four) frames;

Structural frame: Focuses on <u>roles</u> and <u>responsibilities</u> , coordination and control. Organizational charts help define this frame.	Human resources frame: Focuses on <u>providing</u> harmony between needs of the <u>organization</u> and needs of <u>people</u> .
Political frame: Assumes <u>organizations</u> are coalitions composed of varied individuals and interest groups. Conflict and power are key issues.	Symbolic frame: Focuses on <u>symbols</u> and meanings related to events. Culture is important.

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Many Organizations Focus on the Structural Frame

- Most people understand what organizational charts are
- Many new managers try to change organizational structure when other changes are needed
- 3 basic organizational structures
 - Functional-
 - project
 - matrix

Basic Organizational Structures

- Organizational structure depends on the company and/or the project.
- The structure helps define the roles and responsibilities of the members of the department, work group, or organization.
- It is generally a system of tasks and reporting policies in place to give members of the group a direction when completing projects.
- A good organizational structure will allow people and groups to work effectively together while developing hard work ethics and attitudes.
- The four general types of organizational structure are functional, divisional, matrix and project-based.

Basic Organizational Structures

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- **Functional Structure** - People who do similar tasks, have similar skills and/or jobs in an organization are grouped into a functional structure. The advantages of this kind of structure include quick decision making because the group members are able to communicate easily with each other. People in functional structures can learn from each other easier because they already possess similar skill sets and interests.
- **Divisional Structure** - In a divisional structure, the company will coordinate inter-group relationships to create a work team that can readily meet the needs of a certain customer or group of customers. The division of labor in this kind of structure will ensure greater output of varieties of similar products. An example of a divisional structure is geographical, where divisions are set up in regions to work with each other to produce similar products that meet the needs of the individual regions.

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Basic Organizational Structures

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- **Matrix Structure** - Matrix structures are more complex in that they group people in two different ways: by the function they perform and by the product team they are working with. In a matrix structure the team members are given more autonomy and expected to take more responsibility for their work. This increases the productivity of the team, fosters greater innovation and creativity, and allows managers to cooperatively solve decision-making problems through group interaction.
- **Project Organization Structure** - In a project-organizational structure, the teams are put together based on the number of members needed to produce the product or complete the project. The number of significantly different kinds of tasks are taken into account when structuring a project in this manner, assuring that the right members are chosen to participate in the project.

Basic Organizational Structures

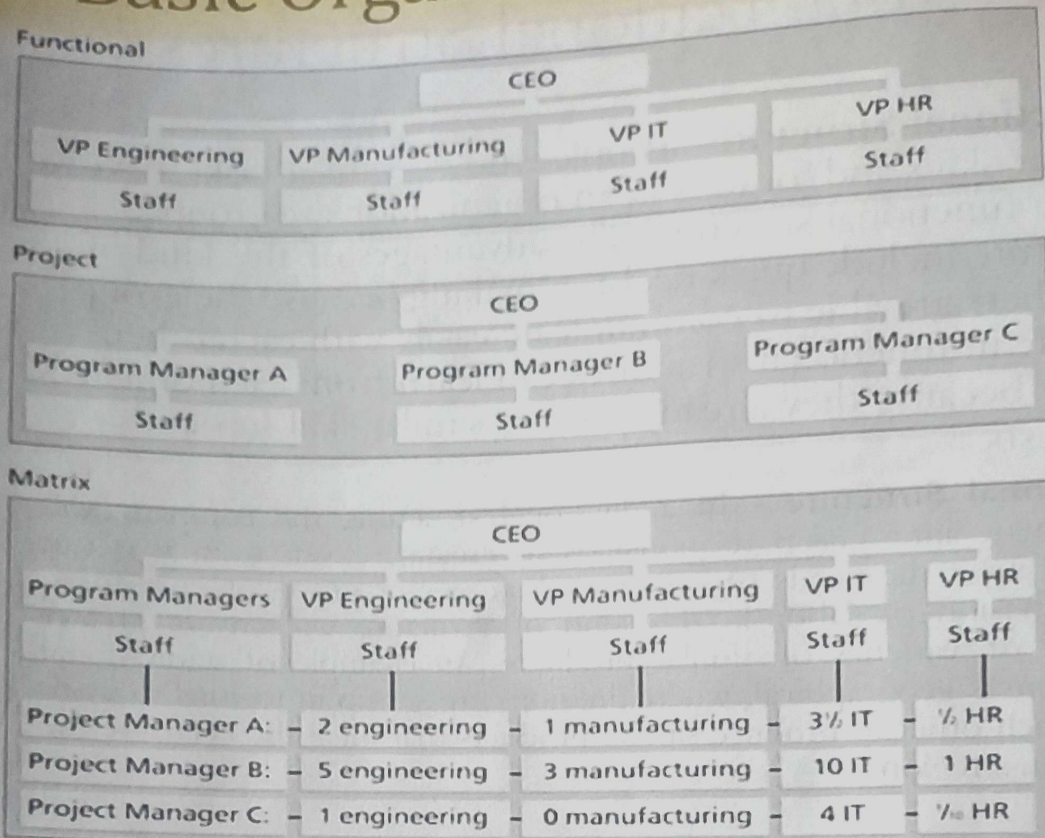


Figure 2-2. Functional, Project, and Matrix Organizational Structures

TECHNICAL TERMINOLOGY:

- A **COMPUTER PROGRAM** is a set of instructions that tells a computer what to do.
- Programs are also called a **SOFTWARE**; software comes in two broad categories.
- **SYSTEM SOFTWARE** describes the programs that operate the computer like operating systems.
- **APPLICATION SOFTWARE** describes the program that allows the user to complete tasks such as creating documents, calculating paychecks and playing games.

TECHNICAL TERMINOLOGY:

- **HARDWARE** are physical devices that make up a computer system.
- **MACHINE LANGUAGE** most basic circuitry level language that computers use to control the operation of the switches represented by 1 and 0.
- **HIGH LEVEL PROGRAMMING LANGUAGE** allow you to use limited keywords that's serves as an identifier that performs task like read, write and add.
- **SYNTAX** or rules of the language with limited vocabulary along with set of rules on using that vocabulary.

TECHNICAL TERMINOLOGY:

- **COMPILER** is a software use to convert the high level language into a machine code.
- **SYNTAX ERROR** is an error message every time a programmer commits an error.
- **METHODS** provide effective methods for ensuring productivity-enhancing processes and resources, and plan for problems that affect integration environments.
- **SOFTWARE** - Software design, code and unit testing, plans, and testing procedures combined with implemented systems, inform us that the created software is being done correctly. It's key to "peer feedback".

TECHNICAL TERMINOLOGY:

- **SYSTEMS** ensure that software design and development standards are allocated for systems to be developed and documented as ready for the software combination and system integration.
- **INTEGRATION** compasses where applications, programs, firmware and hardware can be merged to work together as one.
- **ARCHITECTURE** is defined as a formal description and representation of a system's structure and behavior.

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SYSTEM INTEGRATION:

- is defined as the process of connecting and bringing all components of a subsystems into one system.
- It make sure that the whole system functions as required. It is process of integrating all the components of a system either physical or virtual components.
- The physical components is the hardware, machine systems, inventory and etc. The virtual components composed of data stored in databases, design and algorithms used in the software or applications.

SYSTEM ARCHITECTURE:

- is the conceptual model that defines the structure, behavior and flow of a system.
- is the abstract, conceptualization-oriented, global, and focused to achieve the mission and life cycle concepts of the system. It also focuses on high-level structure in systems and system elements.
- It addresses the architectural principles, concepts, properties, and characteristics of the system-of-interest. It may also be applied to more than one system, in some cases forming the common structure, pattern, and set of requirements for classes or families of similar or related systems.

PROGRAM AND PROJECT PLANNING :

- The aim of program and project planning is to provide the required process steps within integration efforts to expand the planning for programs and software design/development.
- This method of preparation would ensure and define successful strategies and outcomes for carrying out software design/development practices, processes, and implementing procedures that support software and system integration activities.

SYSTEM DESIGN:

- The system design method is to analyze customer requirements and develop a software design/development migration plan to define the architecture, components, modules, interfaces, and data needed for a designed system to meet specified needs.
 - The method of designing systems is becoming increasingly relevant as it offers the disciplines required and applied during life cycles of software design/development.
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SOFTWARE REQUIREMENTS:

- Defined and documented software requirements provide a systematic approach to multiple-resource development.
- Functional software interfaces, performance, testing, and manufacturing results with necessary plans, documentation, and procedures become the basis for software design or development.
- This effective method is used for initial software requirements creation and modifications to design baselines

ORIGIN OF INFORMATION SYSTEM PROJECTS :

To understand more the information system flow means you can look into the details of an IS project. A new or changed IS development projects come from problems, opportunities, and directives and are always subject to one or more constraints.



ORIGIN OF INFORMATION SYSTEM PROJECTS :

- Problems may either be current, suspected, or anticipated. These are undesirable situations that prevent the business from fully achieving its purpose, goals, and objectives (users discovering real problems with existing IS).
- Opportunities are chances to improve the business even in the absence of specific problems. This means that the business is hoping to create a system that will help it with increasing its revenue, profit, or services, or decreasing its costs.
- Directives are new requirements that are imposed by management, government, or some external influence i.e. are mandates that come from either an internal or external source of the business.

ORIGIN OF INFORMATION SYSTEM PROJECTS :

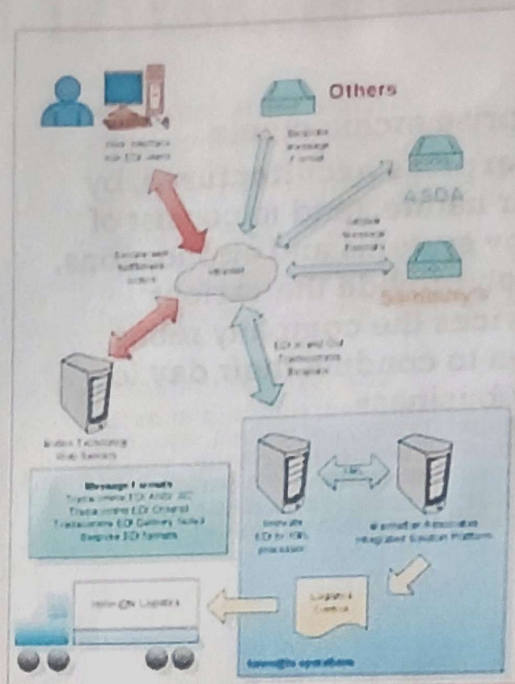
- In working on a project, it must operate in a broad organizational environment and cannot be run isolated. The project managers need to take a holistic or systems view of a project and understand how it is situated within the larger organization.
- One important component to be considered is the stakeholders. These are the people involved in or affected by project activities. Stakeholders include the project sponsor and project team, support staff, customers, users, suppliers, opponents to the project.

ORIGIN OF INFORMATION SYSTEM PROJECTS :

ACCORDING TO THE STANDISH GROUP'S REPORT "CHAOS 2001: A RECIPE FOR SUCCESS," THE FOLLOWING ITEMS HELP IT PROJECTS SUCCEED, IN ORDER OF IMPORTANCE:

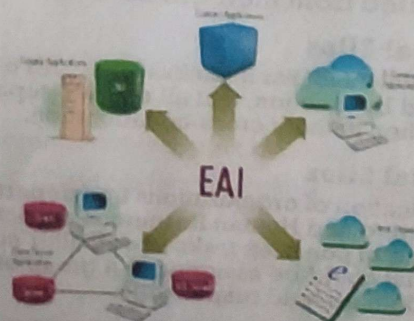
- Executive support
- User involvement
- Experienced project manager
- Clear business objectives
- Minimized scope
- Standard software infrastructure
- Firm basic requirements
- Formal methodology
- Reliable estimates

- Systems integration means that you allow a heterogeneous Information System (IS) to communicate or integrate and share information (or data) seamlessly with one another



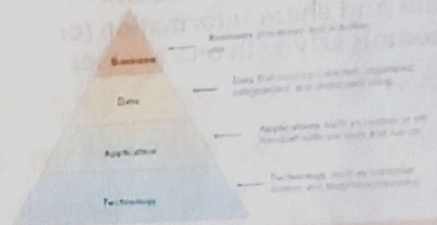
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- In a nutshell, EAI is an approach, or more accurately, a general category of approaches, to providing interoperability between the multiple disparate systems that make up a typical enterprise infrastructure.



BASIC CONCEPTS AND DEFINITIONS

- Enterprise architectures
 - **Enterprise architectures**, by their nature, tend to consist of many systems and applications, which provide the various services the company relies upon to conduct their day to day business.

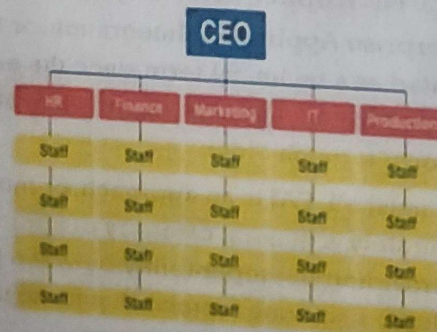


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BASIC CONCEPTS AND DEFINITIONS

- Silos are basically compartmentalized operating units isolated from their environment.
- **Functional Silos**
 - Functional silos in a business are teams of employees, grouped by function, that all operate separately from each other, without cross-collaboration.
- **Horizontal Silos**
 - Classification of organizations into departments like Accounting and Human Resources, reflects the breaking of complex tasks into smaller manageable tasks that could be assigned to a group of people who could then be held responsible.
- **Vertical Silos**
 - Organizations also divided roles in hierarchical layers from strategic planning to management control and operation control

FUNCTIONAL ORGANIZATIONAL STRUCTURE



METHODS ON SYSTEM INTEGRATION

- **Vertical integration**
 - (as opposed to "horizontal integration") is the process of integrating subsystems according to their functionality by creating functional entities also referred to as silos
- **Star integration**
 - also known as spaghetti integration, is a process of systems integration where each system is interconnected to each of the remaining subsystems.
 - When observed from the perspective of the subsystem which is being integrated, the connections are reminiscent of a star, but when the overall diagram of the system is presented, the connections look like spaghetti, hence the name of this method.
- **Horizontal integration**
 - Enterprise Service Bus (ESB) is an integration method in which a specialized subsystem is dedicated to communication between other subsystems.
 - This allows cutting the number of connections (interfaces) to only one per subsystem which will connect directly to the ESB.
- **A common data format**
 - is an integration method to avoid every adapter having to convert data to/from every other applications' formats. Enterprise application integration (EAI) systems usually stipulate an application-independent (or common) data format

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BASIC CONCEPTS AND DEFINITIONS

- **Information Systems in Organizations**
 - Information Systems are a critical component of a successful organization today.
 - Information Systems play a major role in primary and secondary activities of an organization's value chain.
 - The evolution of IS suggests that its role has generally been to support evolving information needs of the organization.
 - Information Systems provide a high level of computer automation to support business functions such as: Accounting, Finance, Marketing, Customer Service, Human Resource Management, Operations, Manufacturing
- **Supporting horizontal silos of organization**
- **Information systems provide analytical and decision making support for management, which is generally categorized into three levels- Strategic, Middle, and Operational.**
- **Supporting vertical silos of organization**

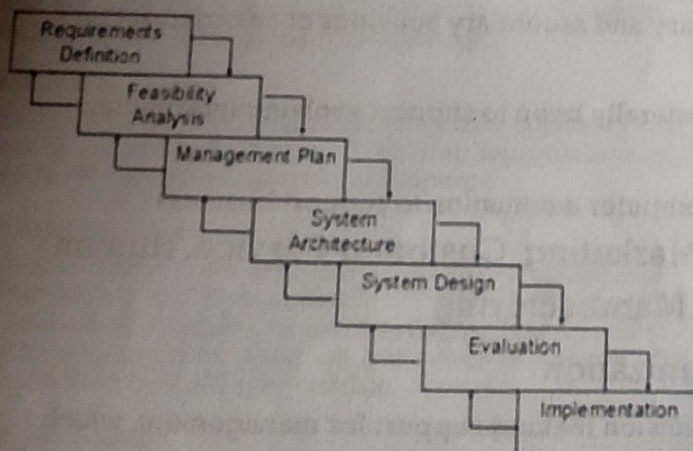
SYSTEM ARCHITECTURE MODEL

- A **system architecture** or **systems architecture** is the **conceptual model** that defines the **structure, behavior, and more views of a system**. An architecture description is a **formal description and representation** of a system, organized in a way that supports **reasoning about the structures and behaviors** of the system.

- A system architecture can consist of system components and the sub-systems developed, that will work together to implement the overall system.

- There have been efforts to formalize languages to describe system architecture, collectively these are called **Architecture Description Languages (ADLs)**.

SYSTEM INTEGRATION LIFE CYCLE



- 1. **Requirement's definition and specification**
- 2. **Feasibility Analysis**
- 3. **System Architecture Development**
- 4. **Management Plan:** program and project plan
- 5. **Systems Design:** logical and physical design
- 6. **Implementation:** design implementation, system tests, and operational deployment
- 7. **Evaluation:** system review and plan for replacement

Figure 1: Systems Integration Life Cycle

SYSTEM INTEGRATION LIFE CYCLE

• SYSTEMS PLANNING

- The objective of the Systems Planning phase is to develop the long-range systems plan for the implementation of an integrated, automated environment within an organization.

• PROJECT INITIATION

- Project Initiation is the first phase in undertaking any of the systems projects identified in the Long-Range Systems Plan.
- Its objective is to lay a proper foundation for the project, by establishing the relationship between the systems project and the Long-Range Systems Plan, by identifying key project personnel, and by establishing that there is adequate funding and commitment.

• ARCHITECTURE DEFINITION

- The objective of the Architecture Definition phase is to confirm and refine the high-level conceptual solution identified during the systems planning phase into an optimum solution to the problem, in the context of the environment, requirements and financial resources available.

• ANALYSIS

- The objective of the Analysis phase is to specify the proposed solution in detail.
- This involves further refining the requirements from the Architecture Definition phase.

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SYSTEM INTEGRATION LIFE CYCLE

• DESIGN

- Basically, the Analysis phase specifies in detail 'what' the system will do. The Design Phase specifies 'how' the application software will accomplish this.

• DEVELOPMENT

- The objective of the Development phase is to construct and test all of the system components. In this phase the system is tested by both developers and users. In addition, all user aids are completed.

• FACILITIES ENGINEERING

- The objective of the Facilities Engineering phase is to create the physical environment including construction of data centers and installation of hardware, and to document this for future reference and expansion needs.

• IMPLEMENTATION

- The objective of the Implementation phase is to make the new system operational. The application software is installed, users are prepared and trained, data files are converted and the application systems and procedures are made operational.

• POST-IMPLEMENTATION SUPPORT

- The objective of the Post-Implementation phase is to maintain and enhance the system to meet the ongoing needs of the user community



SYSTEM ARCHITECTURE AS A PROFESSION

- **Systems architects** are computer professionals, who design, implement, and maintain databases and complex computer systems. They plan projects and activities, solve architectural issues, and maximize quality and efficiency throughout the network or system. Systems architects might work long hours to address database issues without interrupting a company's workday.

Degree Level	Bachelor's degree
Experience	Computer science or software engineering
Licensure and Certification	Voluntary; the Information Technology Infrastructure Library (ITIL) or Project Management Professional (PMP) certification is preferred
Key Skills	Customer service skills; excellent speaking and writing skills; experience with Microsoft Windows and Server software, Linux, TCP/IP and relevant software applications
Salary*	\$106,430 per year (median salary for systems architects in 2016)

https://study.com/articles/Becoming_a_Systems_Architect_Step-by-Step_Career_Guide.html